

StereoWidener Documentation

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September 30, 2026, for StereoWidener 1.0.4 (manual build 6)

1 Introduction

StereoWidener makes your stereo image wider (or narrower) – with seven different algorithms instead of one trick, because a mono synth needs something different from a stereo drum bus or a finished master. Every algorithm has its own interactive display that shows what it does, and the built-in goniometer and level meters show what happens to your signal.

Some of the algorithms are *mono-safe*: the mono sum (L+R) of your signal stays exactly as it was, so your mix still works on a mono speaker. The others create real width from a mono source, but they change the mono sum. StereoWidener tells you which is which, and the Mono Check monitor is one click away.

StereoWidener was made at the Jade Hochschule as a teaching example and a tool for mixing and mastering. To be honest: it is *vibe-coded*. The code was written with an AI (Anthropic's Claude), after an intense planning phase, step by step, with human review, listening and usage tests, and a lot of automated testing. The plans and measurements are part of the source code repository (see section 7).

Have fun!

Joerg aka audio-dsp.

2 Installation

The downloaded file is a standard zip file. It contains a plug-in directory (.vst3), an AU file (.component, macOS only), the Standalone application, the licence files and this manual. Copy the plug-in to the right place for VST3 / AU plug-ins and let your DAW rescan its plug-ins.

2.1 Windows

For Windows™ the standard VST3-directory is:

C:\Program Files\Common Files\VST3

2.2 Mac VST

If you use the VST3 plugin, copy the .vst3 directory to

/Users/yourUSERNAME/Library/Audio/Plug-Ins/VST3

2.3 Mac AudioUnit

If you use the AudioUnit plugin, copy the .component file to
`/Users/yourUSERNAME/Library/Audio/Plug-Ins/Components`

2.4 Linux

For Linux copy the full directory to:

`/home/yourUSERNAME/.vst3/`

2.5 Standalone

The Standalone version runs without a DAW: audio input → StereoWidener → audio output.
Choose your audio device in its settings.

2.6 Preset and settings (De)-Installation

At the first launch of the plug-in the factory presets are copied to the following location (necessary to remove them after you delete the plug-in). The presets are plain XML files.

Windows:

`C:\Users\yourUSERNAME\AppData\Roaming\Jade_Hochschule\StereoWidener`

Mac:

`/Users/yourUSERNAME/Library/Audio/Presets/Jade_Hochschule/StereoWidener`

Linux:

`/home/yourUSERNAME/.config/Jade_Hochschule/StereoWidener`

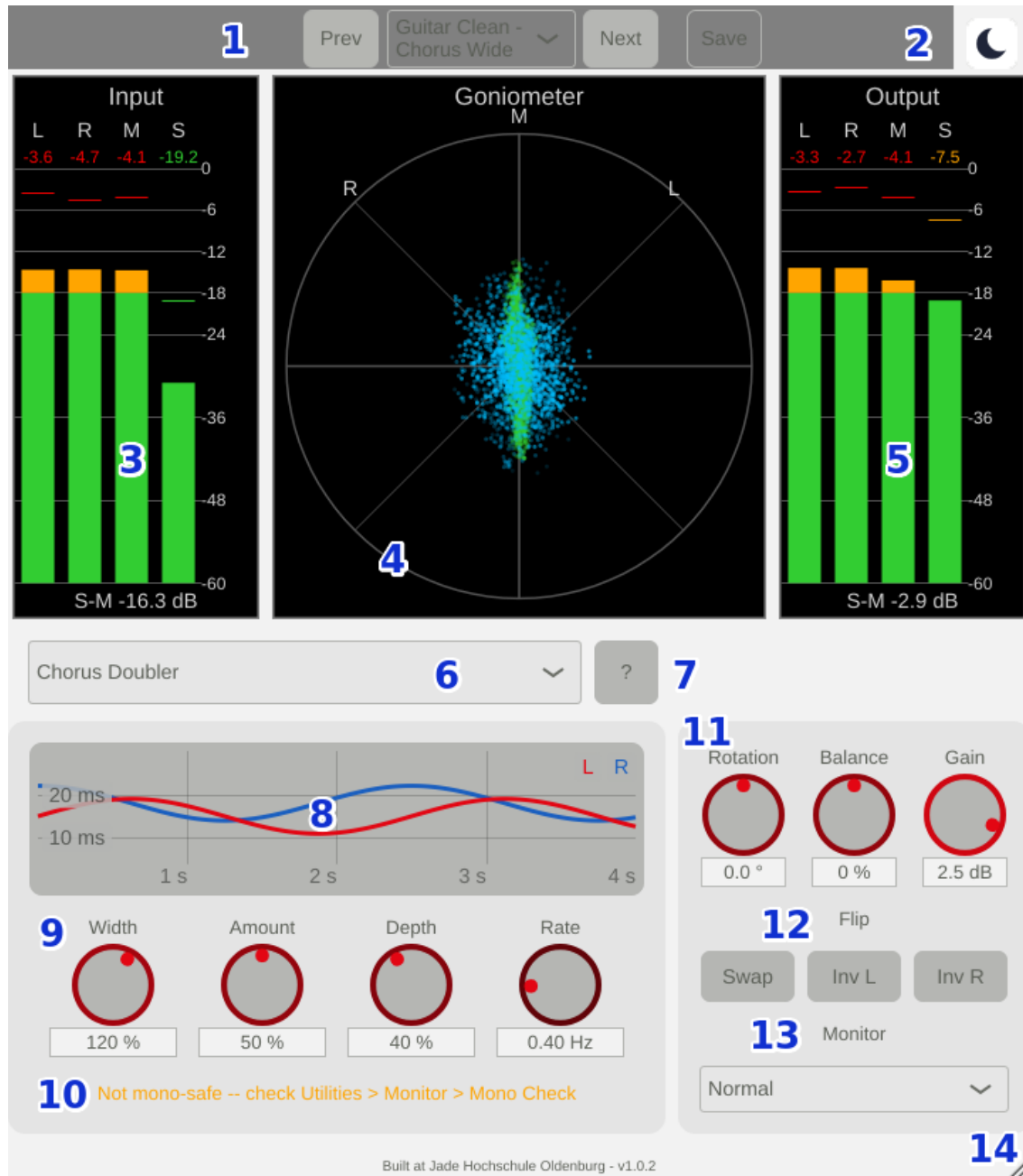
The GUI size, the theme (day/night) and the meter ballistics are stored in a small settings file, `settings.json`:

Windows: `C:\Users\yourUSERNAME\AppData\Roaming\StereoWidener`

Mac: `/Users/yourUSERNAME/Library/StereoWidener`

Linux: `/home/yourUSERNAME/.config/StereoWidener`

3 How to use



StereoWidener consists of five parts: the preset handler on top, the meters, the algorithm selector, the algorithm's own panel (bottom left) and the utilities (bottom right). Signal flow: input → selected algorithm → utilities → output.

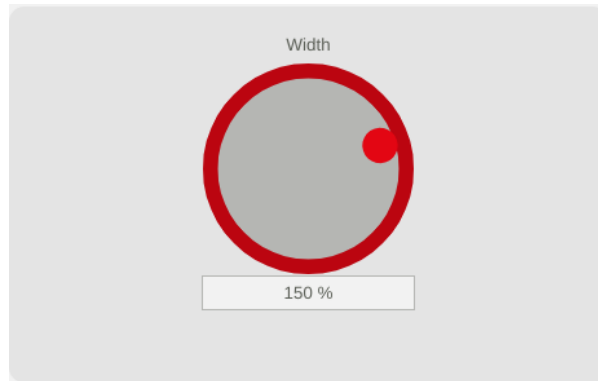
1. **Preset handler:** load and save presets, Prev/Next to step through them. To save a new preset, type a new name and press Save. The Save button turns red if you change any parameter of the displayed preset; press it to update the preset. To delete a preset, delete its XML file (see above).
2. **Theme button:** switches between the night theme (dark) and the day theme (light). It shows the theme a click switches to.

3. **Input meters:** levels of L, R, mid (M) and side (S) of the input signal in dB, the numbers are the peak values. Below: S-M, the level of the side signal relative to the mid signal – the lower, the more mono.
4. **Goniometer:** the stereo image of input (green) and output (blue). A vertical line is mono, a horizontal line is a pure side signal (out of phase), a round cloud is a wide, decorrelated signal. Widening makes the blue cloud wider than the green one.
5. **Output meters:** as (3), for the output signal.
6. **Algorithm selector:** chooses one of the seven algorithms (see section 4). Switching is smooth (crossfaded, no clicks). Each algorithm keeps its own settings.
7. **Help:** a short description of the selected algorithm and all its controls.
8. **Display:** shows what the algorithm does. Most displays are interactive: drag the points, lines or bands, double-click to reset.
9. **Algorithm controls:** the knobs of the selected algorithm. Double-click a knob to reset it to its default (always neutral: no effect).
10. **Mono-safe hint:** appears for the algorithms that change the mono sum.
11. **Rotation, Balance, Gain:** Rotation turns the stereo image (-45° to $+45^\circ$, positive towards R), Balance changes the relative level of L and R (-100 to $+100\%$), Gain is the output level (-24 to $+6$ dB, in 0.5 dB steps).
12. **Flip:** Swap exchanges L and R, Inv L / Inv R invert the polarity of one channel (e.g. to repair a recording with a swapped cable).
13. **Monitor:** Normal; Mono Check (listen to L+R, to hear what a mono speaker plays); Solo Side (listen to the side signal only, to hear what the widening adds). The output Gain is applied after the monitor.
14. **Resize:** drag the corner to change the size of the window.

4 The algorithms

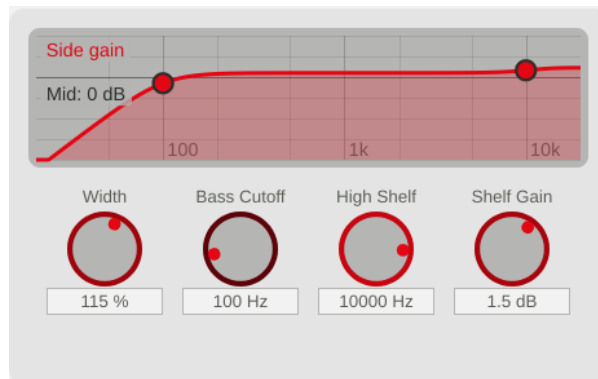
Every algorithm has a **Width** knob: 0% = mono, 100% = unchanged, 200% = double the side signal. All other effect knobs are at “no effect” by default. Rule of thumb: a *mono* source has no side signal, so the M/S algorithms (1, 2, 5) cannot widen it – use an algorithm that creates stereo (3, 4, 6, 7). A *stereo* source is best served by the mono-safe M/S algorithms.

4.1 M/S Width (Broadband) – mono-safe



The classic width knob: the signal is split into mid ($M = (L+R)/2$) and side ($S = (L-R)/2$), the side signal is scaled by **Width**, equally for all frequencies. The mid signal is never changed.

4.2 M/S Width (Filtered / Bass Mono) – mono-safe

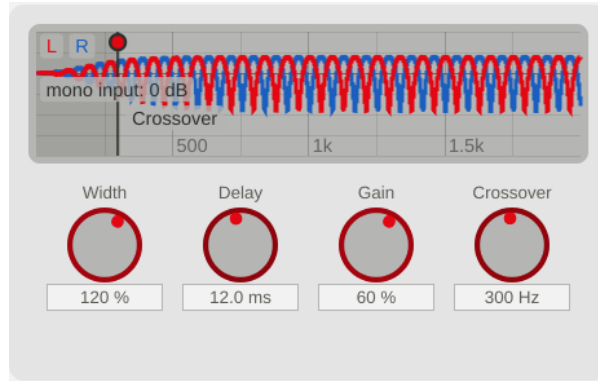


For mastering: like the broadband version, but the side signal is filtered first.

- **Bass Cutoff** (40–500 Hz, Off below 40 Hz): below this frequency the side signal is removed, so the bass stays mono.
- **High Shelf** (1–16 kHz, Off above 16 kHz) and **Shelf Gain** (–6 to +6 dB): raises (or lowers) the side signal above this frequency – a boost adds some “air”.

Display: the gain of the side signal over frequency. Drag the two points for the frequencies, the shelf point up/down for its gain.

4.3 Complementary Comb (Pseudo-Stereo) – mono-safe

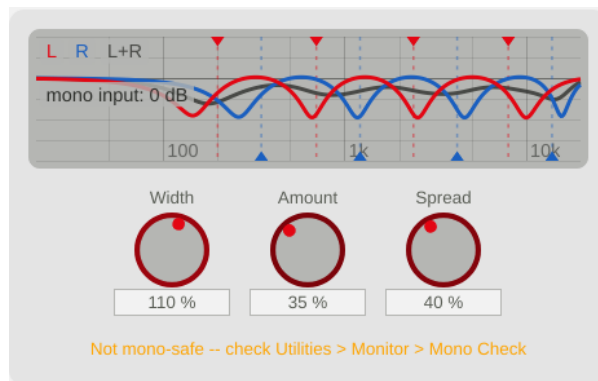


Stereo from mono with a perfect mono sum: a delayed copy of the mid signal is added to the side signal, so L and R get complementary comb filters (the peaks of L are the notches of R). Great for mono synths and electric pianos. On headphones, each channel on its own can sound slightly coloured.

- **Delay** (5–20 ms): spacing of the combs.
- **Gain** (0–100 %): depth of the combs, i.e. the amount of the effect.
- **Crossover** (50–2000 Hz): the effect starts above this frequency, the bass stays untouched.

Display: L (red) and R (blue) for a mono input, 0–2 kHz. Drag the vertical line to move the crossover.

4.4 Allpass Decorrelation

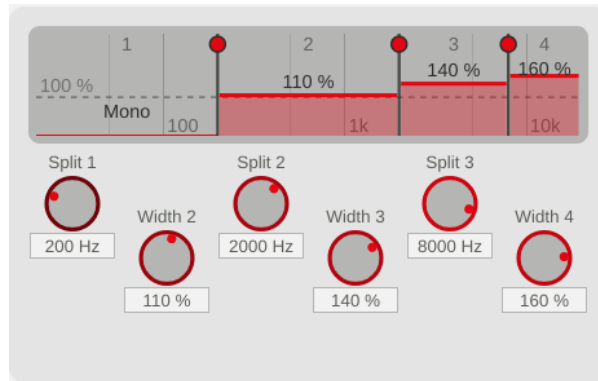


Stereo from mono without audible delay: L and R are blended with two differently phase-shifted copies of the signal. Good for pads, strings and percussion.

- **Amount** (0–100 %): how much of the phase-shifted copies is blended in.
- **Spread** (0–100 %): how different the two copies are (0 % = no decorrelation).

Display: the gain of L, R and the mono sum (grey) for a mono input. Drag left/right for Spread, up/down for Amount. Not mono-safe: the mono sum is coloured – keep Amount moderate (below about 35 %) and check in mono.

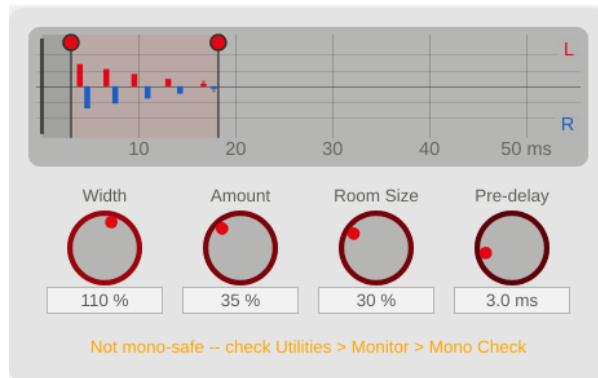
4.5 Multiband Width – mono-safe



Four frequency bands with their own width: band 1 (the bass) is always mono, bands 2–4 have their own **Width** (0–200 %). The three **Split** knobs set the crossover frequencies (40 Hz–18 kHz; each stays between its neighbours). The width knobs sit between the two splits that bound their band. Good for pads, pianos, drum buses and mastering.

Display: the bands on a frequency axis, the height of each bar is its width (dashed line: 100 %). Drag a band up/down for its width, the lines between the bands sideways for the crossovers.

4.6 Early Reflections (Room Widening)

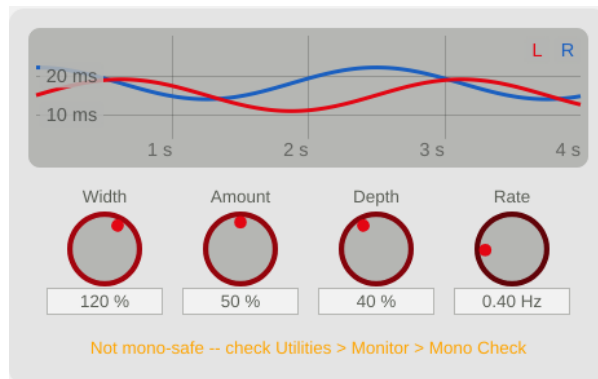


Width from a small virtual room: a few short, quiet reflections, different for L and R. Natural for mono acoustic instruments and vocals.

- **Amount** (0–100 %): level of the reflections.
- **Room Size** (0–100 %): how spread out the reflections are (up to 32 ms).
- **Pre-delay** (0–20 ms): time between the direct sound and the first reflection.

Display: an echogram, L's reflections above the time axis (red), R's below (blue). Drag the two lines for pre-delay and room size, up/down elsewhere for Amount. Not mono-safe.

4.7 Chorus Doubler



Two slowly modulated delay lines, one per channel: the classic stereo chorus, a double-tracking feel for guitars, organs and backing vocals.

- **Amount** (0–100 %): how much of the modulated copies is blended in.
- **Depth** (0–100 %): how far the delay times swing (up to ± 10 ms).
- **Rate** (0.05–2 Hz): how fast they swing.

Display: the delay times of L and R over four seconds. Drag up/down for Depth, left/right for Rate. The least mono-safe algorithm: the mono sum “flanges” – check in mono.

5 The presets

The 20 factory presets are starting points, mostly for one instrument each, using the algorithm that suits it. Presets that change the loudness compensate with the output Gain, so you can compare them fairly. “Init” resets everything to neutral.

Preset	Algorithm	Source
Synth Lead – Pseudo Stereo	Complementary Comb	mono
Synth Bass – Wide Top	Complementary Comb	mono
Synth Pad – Big	Multiband Width	stereo
Synth Arp – Shimmer	Allpass Decorrelation	mono
Guitar Clean – Chorus Wide	Chorus Doubler	mono
Guitar Rhythm – Double	Chorus Doubler	mono
Acoustic Guitar – Mono Mic	Early Reflections	mono
Acoustic Guitar – Stereo Pair	M/S Width (Filtered)	stereo
Piano – Stereo	Multiband Width	stereo
Electric Piano – Pseudo Stereo	Complementary Comb	mono
Organ – Rotary Feel	Chorus Doubler	mono
Strings – Mono Patch	Allpass Decorrelation	mono
Lead Vocal – Space	Early Reflections	mono
Backing Vocals – Wide	Chorus Doubler	mono or stereo
Drums – Overheads	M/S Width (Filtered)	stereo
Drums – Bus	Multiband Width	stereo
Percussion – Mono Shaker	Allpass Decorrelation	mono
Bass Guitar – Grit Only	Complementary Comb	mono
Master – Gentle Widen	M/S Width (Filtered)	full mix
Master – Tight Low End	Multiband Width	full mix

A preset stores all parameters, including the utilities. Factory presets are copied to your preset folder whenever they are missing – a deleted factory preset comes back at the next start. They never overwrite a preset you saved yourself, even under the same name.

6 Tips

- Width is easy to overdo. Switch the plugin off and on (bypass in your DAW) to compare at the same loudness.
- Always check in mono (Monitor: Mono Check), especially with the algorithms that are not mono-safe.
- Keep the bass mono: most algorithms leave the low frequencies alone or have a crossover or cutoff for it.
- Solo Side lets you hear exactly what the widening adds.

7 Source code

This plug-in is open source. You can find the source code, the plans, the documentation of every algorithm and the measurements here:

(https://github.com/JoergBitzer/stereo_widening)

If you are familiar with JUCE, you will have no difficulty understanding what's going on. Please report issues at the GitHub repository.

8 Legal stuff

For this documentation the CC-BY 4.0 licence is valid. Copyright holder is J. Bitzer. The source code of the plug-in is under the MIT licence. The plug-in binaries contain JUCE (used under the AGPLv3), the VST3 SDK (MIT) and, on macOS, the Audio Unit SDK (Apache 2.0); the binaries as a whole are distributed under the AGPLv3.

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